Ankur Pal

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Professional Summary

AI/ML Engineer with 4+ years of hands-on experience developing, training, and deploying machine learning and deep learning models across computer vision (YOLO, CNNs, real-time video analytics), NLP, and Generative/Agentic AI. Experienced in the full ML lifecycle – data preparation and EDA, model development, validation, error analysis, monitoring, and cloud deployment on AWS and GCP through REST APIs and Dockerized microservices. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, NumPy, and Pandas, with production deliveries for international clients in Australia, Brazil, Saudi Arabia, and Japan that cut manual effort by 35-70%.

Technical Skills

Programming & Data: Python, SQL, NumPy, Pandas, Scikit-learn, OpenCV, Exploratory Data Analysis (EDA), Statistical Analysis, Data Preprocessing, Feature Engineering, ETL Pipelines, Data Visualization

Machine & Deep Learning: PyTorch, TensorFlow, ONNX / ONNX Runtime, CNNs, ResNet, EfficientNet, Transformer Architectures, Hugging Face (Transformers, PEFT, TRL), Unsloth, Transfer Learning, LoRA, QLoRA, Supervised Fine-Tuning (SFT), Quantization, Hyperparameter Tuning, Model Evaluation & Benchmarking, Model Monitoring, Inference Optimization

Computer Vision: YOLO, Object Detection, Image Classification, Face Detection & Recognition (YuNet, InsightFace), Facial Landmark Detection, Biometric & Facial Analysis (FairFace), OCR, Real-Time Video Analytics, Vision Language Models (LLaVA, Qwen-VL)

NLP & Generative AI: NLP, Large Language Models (LLM), RAG, Agentic AI, AI Agents, Multi-Agent Systems, Tool Calling, Prompt Engineering, LangChain, LangGraph, LlamaIndex, Sentence Transformers, Embeddings, Semantic Search, Claude API, OpenAI, Groq

Cloud, Deployment & MLOps: AWS (EC2, S3), GCP (Compute Engine, L40 GPU Instances), Azure (Document Intelligence, Logic Apps), Docker, FastAPI, Django, REST APIs, Microservices, CI/CD, Git, Linux

Workflow Automation: n8n, Microsoft Power Automate, Webhook-Driven & Scheduled Pipelines

Databases: MySQL, MongoDB, ChromaDB, Pinecone (Vector Databases)

Professional Experience

FutureVision Remote

Software Engineer – AI/ML *Jul 2026 – Present*

- Build production computer vision and facial-analysis systems for Australian clients: train and deploy **Hugging Face / PyTorch / ONNX** deep learning models behind high-performance **FastAPI** backends for real-time image analysis, and design **LangGraph** agentic pipelines.
- **Rayted:** Train classification and regression models for skin condition screening (23-class dermatological, 18-metric severity scoring) and facial anomaly detection; **l**andmark extraction to compute per-region facial structure metrics for alignment and symmetry scoring.
- Implement end-to-end biometric AI pipelines using **a**nd **m**odels for facial morphometry, biological age/gender/ethnicity estimation, and multi-angle face analysis.

HanuAI Private Limited Mohali, India

Software Engineer – AI/ML *May 2023 – Jul 2026*

- **Road Asset Detection (NHAI):** Engineered an end-to-end **computer vision** system using **YOLO** and Transformer-based architectures to detect road signs, anomalies, and GPS-tagged assets across 500+ km of highway data; trained on 15,000+ custom-annotated images to reach 92% mAP (18% above baseline). Built automated ETL pipelines for ingestion and preprocessing and directed data cleaning and annotation across varied conditions, reducing manual survey effort by 60%.
- **Furniture Anomaly Detection:** Developed an **object detection** product with 90% precision, deployed as a **Dockerized microservice** across 5 client environments on HanuAI's core monitoring platform.
- **Driver Behaviour Detection:** Built a real-time **CNN**-based model with 95% accuracy for fatigue and distraction detection; implemented model monitoring and alerting in production, cutting manual monitoring by 70% and operational costs by 35%.
- **Number Plate Recognition (ANPR):** Deployed a real-time **YOLO + OCR** system at 98% accuracy and 30 FPS on live video streams as a **Docker container on AWS EC2**, with inference optimization for real-time throughput, automatic plate blurring for privacy compliance, and REST API database integration for lookup, logging, and alerts.
- **VLM-Based Error Analysis:** Analyzed false positive and false negative detections and automated their filtering using Vision Language Models (**LLaVA, Qwen-VL**), exposed via REST API endpoints, reducing manual QA intervention and improving pipeline precision.
- **Fine-Tuning & Benchmarking Framework:** Fine-tuned detection and language models with **LoRA, QLoRA, and transfer learning**; built a modular MLOps benchmarking framework (precision, recall, F1-score, inference speed) with **CI/CD hooks for automated evaluation**, adopted across 8 projects and cutting evaluation time by 40%.

- Deployed models on **AWS (EC2, S3)** and **GCP (L40 GPU Instances)** with REST API and Django-based web integration for scalable real-time inference; managed SFTP server setup for client data pipelines.
- Designed **n8n** workflows (REST APIs, webhooks, scheduled triggers) and **Microsoft Power Automate** flows (SharePoint, Teams, Outlook, external APIs) to automate multi-step data pipelines, notifications, and approvals, reducing manual intervention.
- Led end-to-end delivery of AI/ML projects for international and government clients, coordinating cross-functional teams of 30+ (engineers, GIS analysts, developers, field staff) using Agile/Scrum with JIRA and Trello; served as primary technical point of contact from requirements to production deployment.

Soltaire Infosys Private Limited India
Data Science Intern *Jan 2023 – May 2023*

- Performed data preprocessing, **exploratory data analysis (EDA)**, and statistical analysis on real-world datasets; built predictive models (regression, classification, clustering) using **Scikit-learn, Pandas, and NumPy**.
- Assisted in building and evaluating ML pipelines – feature engineering, model training, hyperparameter tuning, and performance benchmarking – and developed data visualization dashboards to communicate insights to stakeholders.

Projects

Rayted | *Python, APIs, Langchain, Langgraph, PyTorch, ONNX, EfficientNet-B0, ResNet-34, YuNet, InsightFace*

- Building an end-to-end AI facial analysis platform that detects, aligns, and profiles faces across demographics, skin health, and dermatological risk in a single pipeline
- Integrating YuNet (CNN face detector trained on WIDER Face) for real-time detection and 5-point landmark extraction, used to align crops for all downstream models
- Deploying a FairFace ResNet-34 multi-task model and an InsightFace demographic CNN for joint age, gender, and ethnicity inference across 18+ output logits
- Fine-tuning EfficientNet-B0 classifiers on Dermnet (23-class dermatological conditions) and a custom 5-class skin-issue dataset for automated screening
- Designing a multi-label EfficientNet-B0 regression model predicting 18 skin severity scores (0-5 scale) spanning acne, hydration, elasticity, and pigmentation
- Langchain/Langgraph agent layer to aggregate multi-model outputs into personalized, human-readable skincare recommendations via API

AI Research Assistant | *RAG, Agentic AI, Claude API, FastAPI, React, ChromaDB, Docker*

- Architected and built a full-stack Agentic RAG system from scratch, combining Retrieval-Augmented Generation with an autonomous multi-step AI agent capable of planning, tool selection, and self-correction — implementing all four pillars of modern GenAI: LLMs, RAG, Agentic AI, and structured generative outputs.
- **RAG Pipeline:** Implemented a complete NLP and indexing pipeline — PDF/text ingestion, ETL preprocessing, recursive text chunking (500 chars, 50-char overlap), dense vector embedding via all-MiniLM-L6-v2 (Sentence Transformers), and cosine similarity search over ChromaDB with 2,000+ indexed chunks; delivered grounded, citation-backed answers from private document collections.
- **Agentic Layer:** Built a multi-iteration think-act loop powered by Claude Sonnet via the Anthropic SDK, leveraging Claude's native tool-use API — the agent autonomously selects between search docs, search web, and save note tools across up to 6 planning iterations, inspects results, and retries when information is insufficient.
- **Structured GenAI Outputs:** Engineered a ReportGenerator module using the Claude API to produce typed JSON outputs — bullet summaries, multi-section research briefs, and side-by-side comparison tables — with robust JSON parsing and markdown rendering.
- **Full-Stack Deployment:** Deployed a FastAPI REST API backend with microservice endpoints for document indexing, RAG Q&A, agent orchestration, and report generation; built a React chat UI with source citation cards showing similarity scores, tool-call badges, animated typing indicators, and a structured report renderer; containerized with Docker for reproducible deployment.

RFP Automation & Subcontractor Management System | *Power Automate, Azure Document Intelligence, Egnyte*

- Designed and delivered an end-to-end RFP processing automation system for a client using Microsoft Power Automate, replacing manual document handling with a fully automated 9-step pipeline covering document ingestion, AI parsing, calendar scheduling, subcontractor matching, approval workflows, and bid tracking.
- Implemented an Egnyte-triggered scheduled automation (hourly polling) that fires when a new RFP folder is created — Power Automate then generates an AI-powered summary of RFP requirements and auto-emails it to pre-configured stakeholder addresses.
- Integrated Azure Document Intelligence (replacing Egnyte parser) for high-accuracy extraction of all deadline dates from RFP documents — automatically translated into calendar invites, eliminating manual date tracking.
- Built a subcontractor matching engine within Power Automate — extracted subcontractor requirements from RFP documents and performed automated lookup against a client-provided subcontractor database, surfacing matched candidates with relevant requirements for approval.
- Implemented a human-in-the-loop approval layer where a stakeholder reviews the matched subcontractor list and requirements before Power Automate dispatches customized outreach emails to all approved subcontractors.

- Automated bid tracking by configuring Power Automate to capture incoming subcontractor responses and update a live Excel sheet in real time as bids arrive — giving the client a single source of truth for bid management without any manual data entry.

AI-Powered Financial Document Analysis (RAG System) | *LangChain, Pinecone, Groq, Streamlit*

- Built an end-to-end RAG pipeline with NLP preprocessing to query financial PDF documents (NVIDIA 10-K) using natural language, enabling instant extraction of financial metrics and insights.
- Implemented ETL data pipeline using LangChain — PDF loading, recursive text chunking (1000 tokens, 100 overlap), and embedding via HuggingFace Sentence Transformers (all-MiniLM-L6-v2).
- Stored and indexed 659+ dense vectors (384 dimensions, cosine similarity) in Pinecone vector database for fast semantic search with top-k similarity retrieval.
- Integrated Groq LLaMA 3.3 (70B) as the LLM backbone with temperature=0 for factual accuracy, chained via LangChain's RetrievalQA for context-aware answer generation.
- Deployed an interactive Streamlit web interface backed by a REST API for real-time document querying; designed scalable pipeline supporting multiple PDF ingestion without re-indexing existing data.

Road Asset & Anomaly Detection with GPS Coordinate Extraction | *YOLO, CNN, Python, OpenCV*

- Built an end-to-end YOLO-based pipeline with automated data pipelines to detect and classify road assets and anomalies from continuous video feeds, extracting precise GPS coordinates via image processing for each detected object.
- Trained on 15,000+ custom-annotated images achieving 92% mAP — 18% above baseline — covering 300 km of highway data; applied inference optimization to meet real-time throughput requirements.
- Integrated a driver drowsiness detection module with 95% accuracy using CNN-based facial landmark analysis for real-time fatigue alerting during road surveys.

Automatic Number Plate Recognition (ANPR) System | *YOLO, OCR, AWS EC2, Docker, REST API*

- Developed a real-time ANPR system using YOLO-based object detection combined with Computer Vision and NLP-based OCR to extract and parse vehicle plate text from live video streams across multiple regions.
- Achieved 98% detection accuracy at 30 FPS in a live production environment through inference optimization, handling varied lighting conditions, angles, and plate formats.
- Integrated automatic number plate blurring for privacy compliance and connected to a backend database via REST API for real-time vehicle lookup, logging, and alert generation on AWS EC2; deployed as a Docker container.

Django-Based QA Automation Dashboard | *Django, REST API, AWS EC2, Docker*

- Designed and developed a full-stack internal QA automation dashboard using Django with REST API endpoints, enabling the team to review, validate, and manage model inference outputs through a structured web interface.
- Deployed on AWS EC2 as a containerized microservice using Docker, with real-time data filtering, annotation review workflows, and role-based access control for QA teams.
- Reduced manual QA review time by centralizing model monitoring and output validation into a single platform, improving overall team efficiency.

Model Fine-Tuning, Benchmarking & Accuracy Optimization Framework | *LoRA, QLoRA, MLOps, CI/CD*

- Fine-tuned object detection and language models using LoRA, QLoRA, and transfer learning techniques with inference optimization to improve domain-specific accuracy on road and asset datasets.
- Built a modular MLOps benchmarking framework to evaluate models across precision, recall, F1-score, and inference speed — adopted as the internal standard across 8 projects; integrated CI/CD hooks for automated model evaluation on new data.
- Reduced model evaluation time by 40% through automated data pipelines and reporting, enabling faster data-driven model selection for production deployment.

VLM-Based False Positive/Negative Automation Pipeline | *LLaVA, Qwen-VL, REST API*

- Integrated Vision Language Models (VLM) — including LLaVA and Qwen-VL — into an automated NLP and vision pipeline to review and filter false positive and false negative detections from the road asset detection system.
- Reduced manual human intervention in model monitoring and output validation by automating edge-case review using VLM reasoning over flagged detection images.
- Improved overall pipeline precision by reducing the false positive rate, directly increasing client confidence and reducing post-processing overhead.

Education

Punjab Technical University — Punjab, India
Bachelor of Technology – Computer Engineering — *Jan 2019 – May 2023*

- CGPA: 7.8 / 10